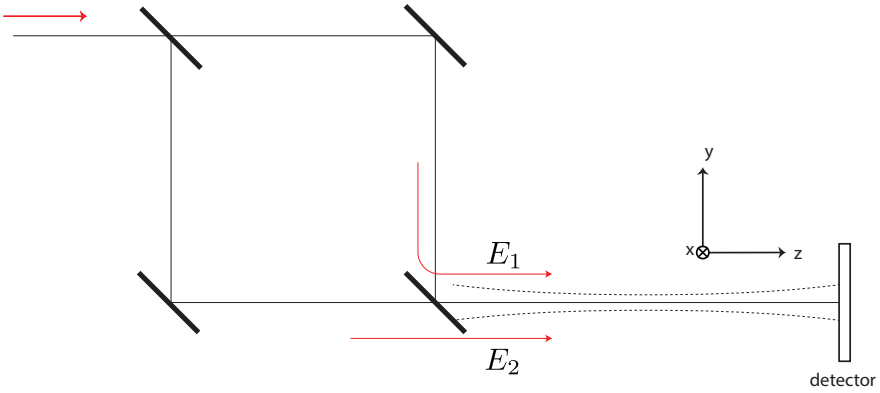
Quantum Optics – Problem Set No. 2

(2.0) Gaussian beam exercises from Problem Set No. 1

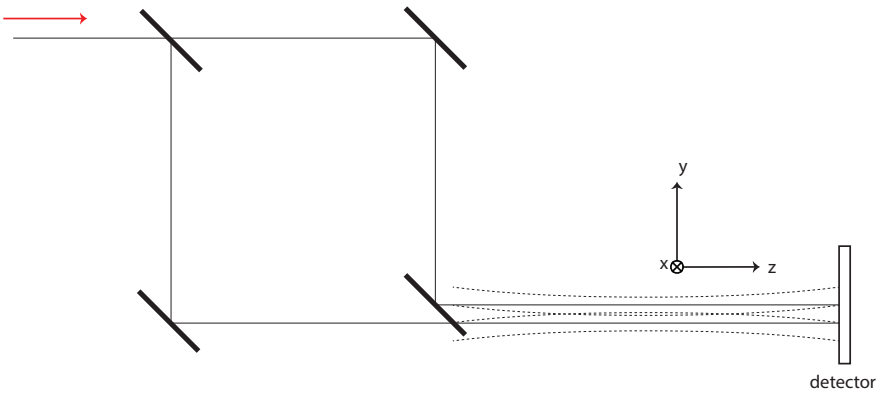
If not yet done, do the exercises (1.3) and (1.4) from Problem Set No. 1.

(2.1) Interference of Gaussian beams

(a) A Gaussian beam enters an interferometer shown below. The interferometer is symmetric such that two identical Gaussian beams $E_1(r,z,t) = E_2(r,z,t)$ are overlapped after the second beam splitter. The beams enter a detector that measures the light intensity integrated over the large detector surface in the xy-plane. If one of the interferometer path lengths is changed by δL, the field E_2 experiences an additional phase shift as $E_2(r,z,t)\exp(-\mathrm{i}k\delta L)$. We assume that δL is small compared to the Rayleigh lengths of the beams and therefore $E_2(r,z,t) \sim E_2(r,z+\delta L,t)$. Calculate the detector signal as a function of δL.



(b) Now the two beams enter the detector with a transversal displacement of dy as shown below. Derive the expression for the detector signal. How does this misalignment affect the interference signal observed at the detector? Does the signal depend on the z-position of the detector?



Hint: $\int_{-\infty}^{\infty} \exp(-ax^2+bx+c)\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}\exp\left(\frac{b^2}{4a}+c\right)$

(2.2) Beam parameter product and uncertainty relation

(a) The divergence (half-)angle θ of a Gaussian beam is given by

$$\theta = \lim_{z\to\infty} \frac{w(z)}{z} \tag{1}$$

with the beam radius $w(z)$. The beam parameter product (BPP) defined as

$$\mathrm{BPP} = \theta\, w_0, \tag{2}$$

where $w_0 = w(0)$ is the waist radius, can be used as a measure how small the beam can be focused. Evaluate the BPP of a Gaussian beam. How does the BPP correspond to the uncertainty relation of a Gaussian beam discussed in the lecture?

(b) $w(z)$ gives the $1/e^2$ intensity radius of the Gaussian beam. When considering more complicated beam shapes, the second-order moment of the intensity distribution is a useful definition of the beam radius[1]. The second-moment radius for the x-direction is defined as

$$\sigma_x(z) = 2\sqrt{\frac{\iint |E(x,y,z)|^2 x^2 dx dy}{\iint |E(x,y,z)|^2 \mathrm{d}x\, \mathrm{d}y}}. \tag{3}$$

Evaluate the BPP of a Gaussian beam using $\sigma_x(z)$ instead of $w(z)$ for the beam radius in Eq. (1) for θ and $\sigma_x(0)$ instead of $w_0 = w(0)$ for the waist radius in Eq. (2).

(c) The higher-order Gaussian modes in the x-direction (Hermite-Gaussian modes TEM_{m0}) are given by

$$E_{m0}(x,y,z) = E_0 \frac{w_0}{w(z)} H_m\left(\frac{\sqrt{2}x}{w(z)}\right) \exp\left(-\frac{x^2+y^2}{w(z)^2} - \mathrm{i}k\frac{x^2+y^2}{2R(z)} + \mathrm{i}\varphi_{m0}(z) - \mathrm{i}kz\right), \tag{4}$$

where $\varphi_{m0}(z) = (m+1)\,\psi(z)$ is the mode-dependent phase proportional to the Gouy phase $\psi(z) = \arctan(z/z_0)$. Calculate the BPP of the TEM_{m0} modes using $\sigma_x(z)$. What does the result mean for the uncertainty relation of higher-order modes? Show that the BPP is the smallest for TEM_{00} mode (Gaussian beam)[2].

Hints:

- Orthogonality:
$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = \sqrt{\pi} 2^n\, n!\, \delta_{nm}. \tag{5}$$
- Recurrence relation:
$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x). \tag{6}$$

[1] This definition, also called the "D4σ method", is recommended for arbitrary beams according to the ISO Standard 11146. See also `www.rp-photonics.com/beam_radius.html` for a brief summary on the beam radius definition.

[2] The ratio of the BPP for a given beam to the BPP of a Gaussian beam is the beam quality factor of the given beam denoted as M^2.